

Retrieve geometry and statistics

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Find the area of a triangle with base 18 cm and height 11 cm, then find the mean of 18, 22, 25 and 15. Copy or complete the first step.

2. A £640 item is reduced by 12.5%. Find the sale price.

3. A cuboid has volume 756 cm³, length 12 cm and width 7 cm. Find its height.

4. Miro says: Miro records a general target such as 'get better at maths'. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Find the area of a triangle with base 18 cm and height 11 cm, then find the mean of 18, 22, 25 and 15. Copy or complete the first step.

Answer 99 cm² and 20.

2. A £640 item is reduced by 12.5%. Find the sale price.

Answer £560.

3. A cuboid has volume 756 cm³, length 12 cm and width 7 cm. Find its height.

Answer 9 cm.

4. Miro says: Miro records a general target such as 'get better at maths'. Is this correct? Explain.

Answer Use assessment evidence to name the exact concept, method, example and support required next.

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Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Find the area of a triangle with base 18 cm and height 11 cm, then find the mean of 18, 22, 25 and 15.

6. Check this result using an inverse, estimate or known fact: A cuboid has volume 756 cm³, length 12 cm and width 7 cm. Find its height.

2. A £640 item is reduced by 12.5%. Find the sale price.

7. Prove or explain your reasoning: Calculate $2\frac{1}{4}$ divided by 3, then write the answer as a decimal.

3. A cuboid has volume 756 cm³, length 12 cm and width 7 cm. Find its height.

8. Identify and correct this misconception: Miro records a general target such as 'get better at maths'.

4. Solve this independently and show every stage: Find the area of a triangle with base 18 cm and height 11 cm, then find the mean of 18, 22, 25 and 15.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: A £640 item is reduced by 12.5%. Find the sale price.

10. Write and solve a similar question to: A cuboid has volume 756 cm³, length 12 cm and width 7 cm. Find its height.

Teacher answers and checking prompts

1. Find the area of a triangle with base 18 cm and height 11 cm, then find the mean of 18, 22, 25 and 15.
Answer 99 cm² and 20.
2. A £640 item is reduced by 12.5%. Find the sale price.
Answer £560.
3. A cuboid has volume 756 cm³, length 12 cm and width 7 cm. Find its height.
Answer 9 cm.
4. Solve this independently and show every stage: Find the area of a triangle with base 18 cm and height 11 cm, then find the mean of 18, 22, 25 and 15.
Answer 99 cm² and 20.
5. Use a second method or representation for: A £640 item is reduced by 12.5%. Find the sale price.
Answer Expected result: £560. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: A cuboid has volume 756 cm³, length 12 cm and width 7 cm. Find its height.
Answer Expected result: 9 cm. A valid check must be shown.
7. Prove or explain your reasoning: Calculate $2\frac{1}{4}$ divided by 3, then write the answer as a decimal.
Answer $\frac{3}{4}$ or 0.75. The explanation must show why the method works.
8. Identify and correct this misconception: Miro records a general target such as 'get better at maths'.
Answer Use assessment evidence to name the exact concept, method, example and support required next.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: A cuboid has volume 756 cm³, length 12 cm and width 7 cm. Find its height.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

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Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Calculate $2\frac{1}{4}$ divided by 3, then write the answer as a decimal.

6. Create a new example that tests the same idea as: A cuboid has volume 756 cm^3 , length 12 cm and width 7 cm. Find its height.

2. Prove the answer to this question, rather than only calculating it: Find the area of a triangle with base 18 cm and height 11 cm, then find the mean of 18, 22, 25 and 15.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: A £640 item is reduced by 12.5%. Find the sale price.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 9 cm.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro records a general target such as 'get better at maths'.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Calculate $2\frac{1}{4}$ divided by 3, then write the answer as a decimal.
Answer $\frac{3}{4}$ or 0.75. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Find the area of a triangle with base 18 cm and height 11 cm, then find the mean of 18, 22, 25 and 15.
Answer Expected result: 99 cm² and 20. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: A £640 item is reduced by 12.5%. Find the sale price.
Answer Expected result: £560. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 9 cm.
Answer One valid reconstruction is: A cuboid has volume 756 cm³, length 12 cm and width 7 cm. Find its height. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro records a general target such as 'get better at maths'.
Answer Use assessment evidence to name the exact concept, method, example and support required next.
6. Create a new example that tests the same idea as: A cuboid has volume 756 cm³, length 12 cm and width 7 cm. Find its height.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.